Chapter 1: Introduction to the GI tract

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Voiceover: Have you ever realized that at your deepest, most inner core you're not really you. Well, think about it.

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The minute you eat something and swallow it and it passes in through your gastrointestinal tract, that's the external environment.

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That food that you just took in was a part of the outside world, and the tract that it's going to follow in through from your mouth till the point that you expel it in the bathroom is actually not you.

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That's the external environment. How weird is that?

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So in this video, we're going to do an overview of the gastrointestinal tract.

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I'll talk about each of the individual parts and what they're main, overall functions are, and then in subsequent videos I'll go through each of these individual parts

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and give a more detailed explanation of how they do what they do.

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All right, so starting off, of course, the first place our food is ever going to go to is our mouth, or the oral cavity.

Chapter 2: Oral cavity functions

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The main functions that we have achieved in the mouth include chewing, the morcellation of food.

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Also very important is hydrolysis.

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As you might recall from biochemistry, hydrolysis is just the enzymatic digestion or the enzymatic breakdown.

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Where chewing is the physical breakdown, hydrolysis is the enzyme-assisted breakdown of food.

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So as we break down food, the goal here is to make what's called a bolus, just a sphere of digested food

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that can then be swallowed and passed on into our next structure.

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After we swallow the food, where do you think it goes?

Chapter 3: Esophagus transport

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This guy right here, and that's the esophagus, the esophagus.

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This, I think, is one of the more boring parts of the GI tract, because all we do here is just propel our bolus. We just pass it on down to the next guy.

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We don't even really do anything to it. Kind of boring. But the next guy is a little more exciting. Now we get to the stomach.

Chapter 4: Stomach digestion

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A lot of action going on in the stomach, one of my favorite parts of the GI tract.

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The stomach is responsible for multiple things, including churning, which is a lot like chewing, except that there are sort of more dimensions

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2 minutes, 16 seconds

of contraction affecting the food and breaking it down.

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We also have hydrolysis going on here, the enzyme-assisted breakdown of food.

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In addition to that, you can store food in your stomach if it's not time to pass it on to the next component of your GI tract.

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The overall goal here is to make what's called chyme.

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So we take our bolus, and we sort of melt it down, per se, into this more fluid type of substance that we can pass on to our intestines.

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So we're moving on now to our intestines.

Chapter 5: Small intestine absorption

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Starting now from about this point right here, we get our duodenum, and then this kind of circulates around here, and then we end up at this point.

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Everything in between, I'll draw it way out here, I'm just going to group together for right now as the small intestine.

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The small intestine.

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There are three parts to this, and we'll talk about that in a subsequent video, but the main functions that we achieve here are

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hydrolysis, and also the absorption of nutrients.

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Notice this is kind of the first part of your GI tract that you're finally taking in some of the break down food products, and using them for nutrition to make other products in your body.

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Great. Now that we have gone through the small intestine, what do you think shows up next?

Chapter 6: Large intestine function

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Starting from after the small intestine ended, all the way through this lined structure right here, we are going to be passing through the large intestine.

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The large intestine. Do you guys remember the other name for the large intestine? It starts with a C.

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If you said "colon," absolutely right.

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The colon is also one of the more boring parts of the GI tract, because really all we have going on here is absorption, but not of nutrients per se,

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more like things like water, or ions, or vitamin K, just things like that are absorbed in the large intestine, so not a very high yield place for acquiring nutrients.

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Then after that, we're going to pass food on to this structure here. This is called the rectum.

Chapter 7: Rectum and expulsion

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The rectum.

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Kind of like the stomach of the GI tract, the rectum serves for storage, We hold on to our processed food,

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if we can call it that anymore at this point, and it's held there until we deem it an appropriate time to expel the food.

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So when it's time to expel the food, it'll come out through the anus, through expulsion, expulsion.

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So those are all the key components of our gastrointestinal tract.

Chapter 8: Accessory organs overview

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There are some other accessory organs that are involved here in digestion, and I'll have videos that talk about them as well.

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Those include things like the liver, the gallbladder, the pancreas, and those will come up in subsequent videos.